Radiomics for Medical Image Classification

A Hands-on Student Project

Department of Radiological Technology

Machine Learning in Medical Imaging

Outline

1. 🔬 What is Radiomics & Why It Matters
2. 📊 Types of Radiomic Features
3. ⚙️ The Advanced Pipeline
4. 💾 Available Datasets
5. 🚀 Getting Started & Notebooks
6. 📈 Machine Learning Workflow
7. 🏆 Best Practices & Common Pitfalls
8. 📝 Assignment Guidelines

The Clinical Motivation

Two head-and-neck tumors may **look identical** on a CT scan.

But radiomics reveals one has **higher entropy** and **irregular texture**, predicting aggressive behavior.

> *In a study of 300 HNSCC patients, radiomic features predicted HPV status with **AUC > 0.80** — without biopsy.*
> *— Vallières et al., Scientific Reports, 2017*

✅ **Non-invasive** | ✅ **Quantitative** | ✅ **Reproducible**

What is Radiomics?

Definition: **Radiomics** = Extracting **quantitative features** from medical images to:

- Predict disease outcomes
- Classify tumor types
- Personalize treatment

The Pipeline

Radiomics Over Traditional Imaging

Aspect	Traditional Imaging	Radiomics Approach
Assessment	Qualitative	✅ Quantitative
Interpretation	Subjective	✅ Objective
Reproducibility	Low	✅ IBSI-standardized
Prediction	Limited	✅ Predictive
Invasiveness	May need biopsy	✅ Non-invasive

IBSI = Image Biomarker Standardisation Initiative.

Types of Radiomic Features

1. First-Order Statistics (19 features)

- Mean, median, variance, skewness, kurtosis
- Energy, entropy
- **What they capture**: Intensity distribution

2. Shape Features (14–16 features)

- Volume, surface area, sphericity, compactness
- Elongation, flatness
- **What they capture**: Tumor geometry

Texture Features (~70 features)

Matrix	Full Name	What It Measures
GLCM	Gray Level Co-occurrence	Spatial relationships
GLRLM	Gray Level Run Length	Consecutive same-intensity
GLSZM	Gray Level Size Zone	Connected regions
GLDM	Gray Level Dependence	Voxel neighborhood
NGTDM	Neighboring Gray Tone	Intensity differences

Filter-Based Features (Optional)

4. Filter-Based Features

- **Wavelet**: Multi-scale frequency decomposition
- **LoG (Laplacian of Gaussian)**: Edge enhancement

> *"Filters can reveal patterns not visible in the original image intensities."*

⚙ The Project Pipeline

Phase 1: Feature Extraction

```
CT/MRI Image + Segmentation Mask
     ↓
PyRadiomics Library (IBSI-compliant)
     ↓
~100–1500 Features per patient
     ↓
CSV File with features
```

Phase 1b: Advanced Image Processing

Deconstructing the "Black Box"

- **Normalization**: Z-score vs. Min-Max math
- **Binning**: How `binWidth` affects texture matrices
- **Filters**: Visualizing Wavelets and LoG responses

> ***Goal**: Mathematically justify why we choose certain preprocessing steps for certain modalities.*

Phase 2: Machine Learning

```
Extracted Features
     ↓
Preprocessing → Feature Selection
     ↓
Model Training (Cross-Validation)
     ↓
Evaluation & Interpretation
```

Phase 1: PyRadiomics Config

Configuration Example

```yaml
Settings:
  binWidth: 25            # Discretization
  normalize: True         # Normalize intensities

Feature Classes:
  firstorder: True        # 19 features
  shape: True             # 14 features
  glcm: True              # 24 features
```

Phase 2: ML Pipeline Overview

Step 1: Preprocessing

- Handle missing values, **Standardization** (z-score)

Step 2: Feature Selection

- Remove highly correlated features ($|r| > 0.9$)
- Reduce from 100+ to 10–20 informative features

Step 3: Model Training

- 5-fold stratified cross-validation
- Evaluate on held-out test set

💾 Available Datasets

Dataset Options

Option	Source	Difficulty	Format
Synthetic	Auto-generated	⭐ Beginner	NIfTI
radMLBench	GitHub	⭐⭐ Intermediate	CSV
TCIA	Cancer Archive	⭐⭐⭐ Advanced	DICOM

Recommended: radMLBench Pre-extracted Features

radMLBench Datasets

Dataset	Modality	Samples	Features	Task
LNDb	CT	173	105	Lung Nodule
HNSCC	CT	93	105	HPV Status
Head-Neck	CT	137	105	Survival

Download: `pip install radMLBench` or CSV from GitHub.

🚀 Project Structure

```
radiomics_student_project/
├── notebooks/
│   ├── 00_data_exploration.ipynb
│   ├── 01_feature_extraction.ipynb
│   ├── 01b_advanced_image_processing.ipynb # New!
│   └── 02_machine_learning.ipynb
├── scripts/              ← Helper scripts
├── data/                 ← Datasets
...
```

Running the Notebooks

Method	Setup	Best For
Google Colab	Upload → Run	🟢 No install
Local Jupyter	`pip install`	🟡 Full control

Tip: Start with `00_data_exploration.ipynb` to understand your dataset before jumping into ML.

Notebook 01: Extraction

Key Code

```
from radiomics import featureextractor

extractor = featureextractor.RadiomicsFeatureExtractor(**settings)
# enable classes...
result = extractor.execute(image, mask)
```

Output: CSV file ready for machine learning →

Notebook 02: ML Pipeline

ML Pipeline (using sklearn Pipeline)

```
from sklearn.pipeline import Pipeline

pipe = Pipeline([
    ('scaler', StandardScaler()),
    ('selector', SelectKBest(f_classif, k=10)),
    ('model', RandomForestClassifier())
])
cv_scores = cross_val_score(pipe, X_train, y_train, cv=5)
```

Cross-Validation: Why & How

- A single split is **fragile** — results vary.
- Cross-validation uses **all data** for both training and testing.

5-Fold Stratified CV

```
Fold 1: [Test][Train][Train][Train][Train]
Fold 2: [Train][Test][Train][Train][Train]
...
Final Score = Mean(Scores) ± Std
```

Performance Metrics

Metric	When to Use	Clinical Meaning
Accuracy	Balanced classes	Overall correctness
Precision	Avoid False Positives	Avoid over-treatment
Recall	Avoid False Negatives	Find all sick patients
AUC-ROC	Threshold-indep.	Overall discrimination

AUC > 0.8 → Good | **AUC > 0.9** → Excellent

Algorithms to Compare

Model	Strengths
Log. Regression	Simple, baseline
Random Forest	Handles interactions
SVM (RBF)	Non-linear, small samples
SVM (Linear)	Interpretable, fast

Use `GridSearchCV` for hyperparameter tuning.

Feature Importance

Why It Matters

- Understand **what drives** predictions.
- Verify **clinical plausibility**.

Common Top Features

- 📐 **Volume, Sphericity** (Shape)
- 🔢 **Entropy** (First-order)
- 🧩 **Contrast, Correlation** (GLCM)

🛡️ Best Practices

- ❌ **No Data Leakage**: Scale/select only on training data.
- ✅ **Use Pipelines**: `sklearn.Pipeline` prevents leakage.
- ✅ **Correlation**: Remove features with **|r| > 0.9**.
- ✅ **Report**: Use CV mean ± std and held-out test scores.

> ***Rule of thumb***: *With < 100 samples, use ≤ 10 features.*

⚠️ Common Pitfalls

Pitfall	Problem	Solution
Overfitting	Memorizes train data	Regularize, simplify
Leakage	Test info in train	Use `Pipeline`
Imbalance	Biased predictions	Stratified, Weights
Redundancy	High correlation	Correlation removal
Small N	Unstable results	Simpler models

📝 Assignment Guidelines

Level	Task
⭐ **Basic**	Complete NBs with synthetic data
⭐⭐ **Intermed.**	radMLBench, compare 3 models
⭐⭐⭐ **Adv.**	Handle imbalance, ensembles

Submission: Notebooks + Worksheet + 1-page report.

Expected Learning Outcomes

- ✅ **Understand** the radiomics workflow.
- ✅ **Analyze** image processing math (NB 01b).
- ✅ **Extract** features (IBSI-compliant).
- ✅ **Preprocess** without data leakage.
- ✅ **Train** and compare ML models.
- ✅ **Evaluate** with AUC, F1, and confusion matrix.
- ✅ **Interpret** feature importance.
- ✅ **Avoid** common ML pitfalls.

Resources & Key Papers

- **PyRadiomics**: pyradiomics.readthedocs.io
- **Scikit-learn**: scikit-learn.org
- **radMLBench**: github.com/aydindemircioglu/radMLBench

1. Lambin et al. (2012) — Radiomics intro
2. Zwanenburg et al. (2020) — IBSI Standard
3. Vallières et al. (2017) — HPV Prediction

Questions?

Let's Get Started! 🚀

Option A: Start from images → `01_feature_extraction.ipynb`

Option B: Jump to ML → `02_machine_learning.ipynb`

Appendix: Setup Commands

Install Dependencies

```
pip install -r requirements.txt
jupyter notebook notebooks/
```

Download radMLBench (Python)

```
import pandas as pd
url = ".../LNDb.csv.gz"
df = pd.read_csv(url, compression='gzip')
```

Thank You!

Happy Learning 🎓

Contact: anucha.cha@mahidol.ac.th